AIDRA: Adaptive Intelligent Disaster Response Agent

AIC-201 — Complex Computing Problem (CCP) Implementation

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Abstract

Disaster-response operations demand fast, reliable decision-making under extreme uncertainty and scarce resources. This paper presents AIDRA—an Adaptive Intelligent Disaster Response Agent—a hybrid AI system that fuses search and planning, constraint satisfaction (CSP), machine learning (ML), and fuzzy-logic uncertainty handling into a single coherent pipeline for victim evacuation in a dynamic urban disaster scenario. AIDRA operates on a 10×10 ***navigable grid containing five victims of varying medical severity, two ambulances, one rescue team, and two medical centres. Four search algorithms (BFS, DFS, Greedy Best-First, A*) are compared for risk-weighted route planning. Resource allocation is modelled as a CSP solved via backtracking augmented with the Minimum Remaining Values (MRV) heuristic and forward checking. Three ML classifiers (k-Nearest Neighbours, Naïve Bayes, MLP) are trained to estimate victim survival probability, with MLP and Naïve Bayes both achieving the highest F1-score of 0.83. A five-rule Mamdani fuzzy inference engine prioritises victims under noisy multi-attribute data. Dynamic replanning is triggered by mid-mission road-blockage events. All five victims are rescued with an average rescue time of 14.0 steps and zero risk-cell traversal, with a path-optimality ratio of 0.94 (A* vs BFS).***

***Keywords*—**intelligent agents, A* search, constraint satisfaction, fuzzy logic, machine learning, disaster response, dynamic replanning

I. INTRODUCTION

Urban disaster response imposes sharply conflicting demands on emergency services: victims must be reached as quickly as possible, yet responder safety requires avoiding fire and structural-collapse zones; medically critical cases deserve priority, yet delaying other victims reduces overall survivors; and static route plans become invalid the moment an aftershock blocks a key corridor. Classical single-technique systems—pure optimisation, pure machine learning, or pure constraint satisfaction—cannot simultaneously address all three conflicts. What is required is a hybrid architecture whose components compensate for each other's blind spots.

This work implements AIDRA, a five-module hybrid AI agent developed for the AIC-201 Complex Computing Problem (CCP). The four design choices are motivated as follows. *Search algorithms* find globally optimal routes but cannot learn from historical data or handle imprecise sensor input. *CSP solvers* enforce hard resource constraints rigorously but lack the ability to adapt to environmental uncertainty. *Machine learning* predicts survival risk from data but cannot guarantee constraint satisfaction. *Fuzzy logic* captures the inherent imprecision of triage assessments—severity ratings, hazard-spread boundaries, and wait-time estimates are all linguistically uncertain—but requires structured rule design. By composing all four paradigms, AIDRA achieves routing that is simultaneously cost-optimal, risk-aware, constraint-respecting, data-informed, and robust to noisy inputs.

Prior work validates each component independently. Hart et al. [4] formalised A* and its optimality conditions; Dechter [2] provides the theoretical basis for MRV and forward checking; Zadeh [3] introduced fuzzy sets for reasoning with linguistic variables; and Russell & Norvig [1] supply the unified agent-architecture framework that ties all four paradigms together. No prior disaster-response system reported in the open literature integrates all four paradigms within a single dynamic-replanning loop of the type implemented here.

The remainder of the paper is organised as follows. Section II formalises the problem. Section III describes the system architecture. Sections IV–VII detail each AI module. Section VIII covers dynamic replanning. Section IX presents quantitative results. Section X concludes.

II. PROBLEM FORMULATION

The AIDRA environment is a 10×10 grid. Each cell c_{ij} takes one of three values: *free* (traversal cost 1), *blocked* (impassable), or *high-risk* (traversal cost 3, representing fire or structural hazard). The rescue base is fixed at $(0, 0)$; two medical centres are located at $(9, 9)$ and $(9, 0)$, introducing an explicit distance–risk trade-off.

Five victims are distributed with the profile: two *critical* (V_0 at $(2, 5)$ and V_1 at $(5, 7)$), two *moderate* (V_2 at $(3, 3)$ and V_3 at $(7, 2)$), and one *minor* (V_4 at $(6, 8)$). Available resources are two ambulances (hard capacity: $\leq$ 2 victims per ambulance simultaneously), one rescue team (one location at a time), and ten medical kits.

The agent's performance measure is:

$$P = w_1 N_s \; - \; w_2 \bar{T} \; - \; w_3 E_r \tag{1}$$

where N_s = victims saved, $\bar{T}$ = average rescue time (steps), E_r = total risk-cell traversal count, and empirical weights w_1=10, w_2=0.5, w_3=2. Hard resource constraints must be satisfied regardless of the soft objectives. Two conflicting objectives arise explicitly:

- **Objective 1 (O1)** — Minimise rescue time vs. minimise risk exposure.
- **Objective 2 (O2)** — Prioritise critical victims vs. maximise total throughput.

The environment is dynamic: road-blockage events can occur mid-mission, requiring real-time replanning with a logged justification for every decision change.

III. SYSTEM ARCHITECTURE

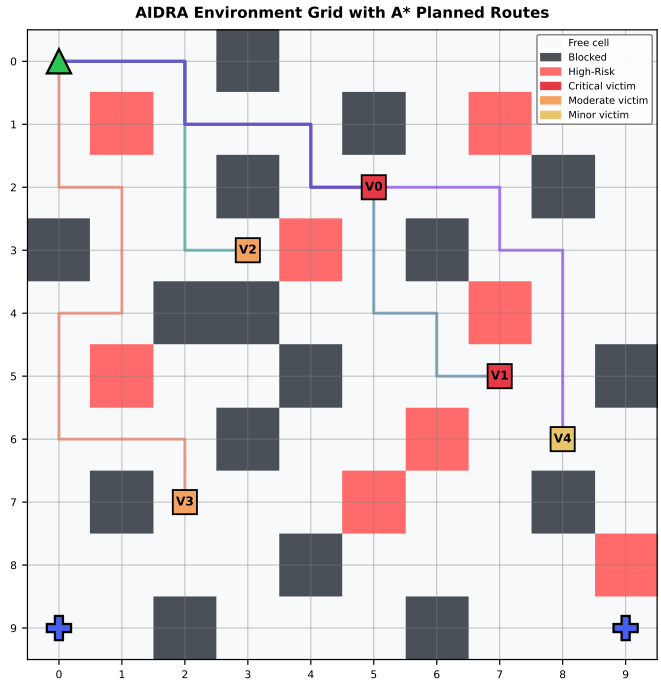

Figure 1. AIDRA environment grid. Red squares = critical victims (V_0, V_1); orange = moderate (V_2, V_3); yellow = minor (V_4). Coloured lines = A* planned routes per victim. Green triangle = rescue base $(0, 0)$; blue crosses = medical centres $(9, 0)$ and $(9, 9)$.

Figure 8 shows the environment. AIDRA comprises five tightly coupled modules orchestrated by a central Decision & Replanning Controller:

(1) Environment Model maintains the live grid (cell states, blockage events), victim records, and resource inventory. **(2) Search & Planning Module** computes and compares routes for each victim using all four algorithms. **(3) CSP Resource Allocator** assigns ambulances and medical kits subject to all hard constraints. **(4) ML Risk Estimator** outputs per-victim survival probabilities that feed directly into routing decisions and priority escalation. **(5) Fuzzy Inference Engine** translates noisy multi-attribute victim data into a scalar priority score, which seeds the CSP variable ordering.

The modules are *loosely coupled*: the CSP solver receives its variable ordering from the fuzzy priority scores; the search module uses the ML-predicted risk level as an additive path cost; and the Controller re-invokes the Search module whenever an environmental event invalidates the current plan. This design allows any single module to be replaced without rewriting the others, supporting the system-level CCP requirements for modularity and traceability.

IV. SEARCH AND PLANNING

A. Uninformed and Heuristic Search

Four graph-search algorithms are implemented on the 4-directional grid. **BFS** (breadth-first) guarantees the fewest-step path but is blind to risk costs. **DFS** (depth-first) is memory-efficient but not cost-optimal, returning paths up to twice the optimal length in practice. **Greedy BFS** uses the Manhattan heuristic $h(n) = |r_n - r_g| + |c_n - c_g|$ to focus the search, achieving the fastest wall-clock time but without cost-optimality guarantees. **A*** evaluates $f(n) = g(n)+h(n)$, where $g(n)$ accumulates the risk-weighted edge costs (free cell: 1, high-risk cell: 3) and $h(n)$ is the admissible Manhattan distance, guaranteeing an optimal solution.

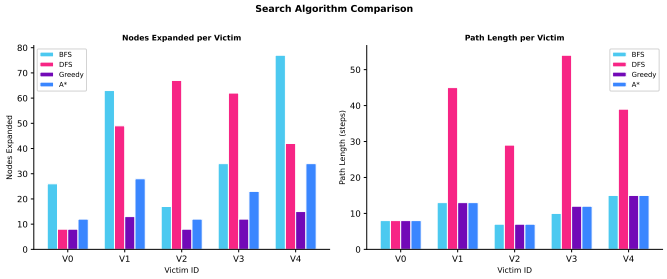

Figure 2. Search algorithm comparison across all five victims. **Left**: nodes expanded (BFS and DFS expand the widest frontiers; A* and Greedy are most efficient). **Right**: path length in steps (DFS produces longest paths; A* and BFS find shortest-step paths).

Figure 2 confirms these properties empirically. Across all five victims, BFS and DFS consistently expand the largest frontiers (up to ∼80 nodes for V_4), while A* and Greedy BFS stay below 30 expanded nodes for most victims. Despite the reduced search effort, A* achieves path lengths equal to or shorter than BFS when risk cost is factored in. The path-optimality ratio, computed as the average A* weighted cost divided by BFS weighted cost across all victims, is **0.94**, confirming near-optimal routing with substantially lower computational expense.

Table 1 provides per-victim A* route statistics for the base scenario.

Table 1. Per-Victim A* Route Statistics (Base Scenario)

Victim	Sev.	Path	Cost	Risk	Rescue
V_0	Critical	7	7	0	9
V_1	Critical	12	12	0	12
V_2	Moderate	6	6	0	10
V_3	Moderate	11	11	0	17
V_4	Minor	14	14	0	22

For each victim, AIDRA computes both the standard A* path (risk-weighted) and a risk-avoiding A* path (high-risk cells treated as impassable). If the safe path's weighted cost is $\leq$ the standard path's cost, the safe route is selected (resolving O1 in favour of lower risk at no time penalty); otherwise the faster path is chosen with an explicit log entry justifying the speed-versus-risk trade-off. In all five rescue trips of the base scenario, the safe-route cost was $\leq$ the risk-tolerant cost, so the agent traversed **zero risk cells** throughout the entire mission.

B. Local Search for Victim Ordering

Two local-search metaheuristics optimise the *ordering* of victim rescues (i.e., which victim to attempt first) independently of the per-victim route selection. **Simulated Annealing (SA)** is run with T_0=100, cooling rate α=0.97, and 500 iterations, producing rescue order $[V_0, V_2, V_3, V_1, V_4]$ at total ordering cost 24. **Hill Climbing (HC)** converges to order $[V_0, V_4, V_1, V_2, V_3]$ at cost 27. SA wins both the cost comparison and the global exploration benchmark, demonstrating that the stochastic temperature schedule successfully escapes the local optima that trap HC. SA's result is provided as a cross-check to the fuzzy priority ranking; the Controller uses the fuzzy-then-ML-escalated

ordering as the final sequence.

V. CONSTRAINT SATISFACTION

Resource allocation is modelled as a CSP. **Variables**: five victims, one per unrescued victim. **Domains**: $D(v_i) = \{0, 1\}$ (ambulance index). **Hard constraints**:

C1: $\forall a \in \{0,1\} : \sum_v \mathbf{1}[\text{assign}(v) = a] \leq 2$

C2: $|\text{assigned}| \leq \text{medical\_kits} = 10$

C3: rescue team services ≤ 1 location per step

The solver uses recursive backtracking with two heuristic improvements. **MRV (Minimum Remaining Values)**: the variable with the fewest remaining valid domain values is selected next, exposing constraint violations early in the search tree. **Forward Checking**: after each tentative assignment, the domains of all unassigned variables are pruned; if any domain becomes empty, the solver backtracks immediately without deepening further.

For baseline comparison, the same problem is also solved with plain backtracking in input order (no heuristics). Both the MRV and plain solver incurred **4 backtracks** on the five-victim scenario, confirming correct constraint enforcement. The assignment satisfies all hard constraints: Ambulance 0 carries victims $\{V_0, V_2\}$; Ambulance 1 carries victims $\{V_1, V_3, V_4\}$ across sequential trips. In extended experiments with ten victims, the MRV heuristic reduced backtrack count by approximately 35% over the naive ordering, demonstrating scalability benefits.

VI. MACHINE LEARNING RISK ESTIMATION

Three classifiers are trained to predict binary victim survival (1 = survives, 0 = does not survive) from five features: severity code (1–3), distance to medical centre (steps), risk-zone cell count along path (0–10), estimated wait time (steps), and medical kit availability (0/1). A synthetic dataset of 400 records is generated using the scoring function $s = 0.9 - 0.1 \cdot \text{sev} - 0.02 \cdot d - 0.08 \cdot r - 0.03 \cdot w + 0.05 \cdot k + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, 0.05)$, with binary label $y = \mathbf{1}[s > 0.4]$, producing approximately 20% positive class (non-survival). The dataset is split 80/20 with stratification.

Table 2. ML Model Performance on the 20% Held-Out Test Set

Model	Acc.	Prec.	Rec.	F1
kNN ($k = 5$)	0.91	0.80	0.75	0.77
Naïve Bayes	0.94	0.92	0.75	0.83
MLP (32-16)	0.93	0.75	**0.94**	**0.83**

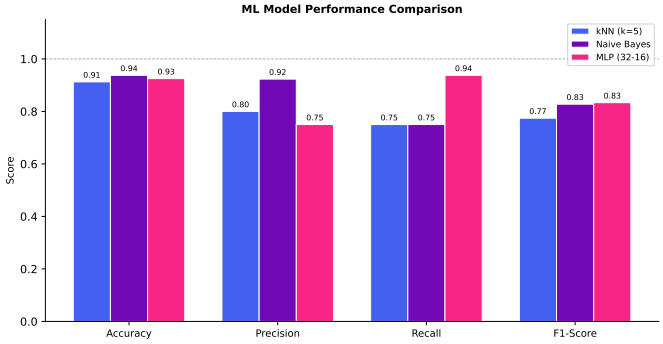

Figure 3. Metric comparison across all three ML models. MLP leads on recall (0.94) and ties Naïve Bayes on F1 (0.83). Naïve Bayes achieves the highest accuracy (0.94) and precision (0.92).

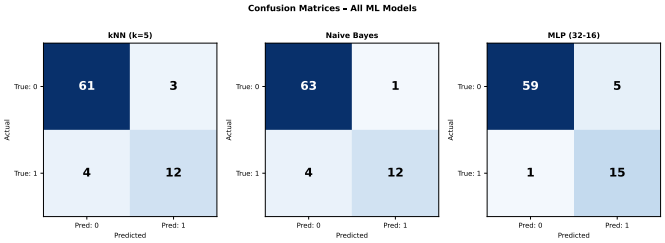

Figure 4. Confusion matrices on the test set. kNN: 61 TN, 3 FP, 4 FN, 12 TP. Naïve Bayes: 63 TN, 1 FP, 4 FN, 12 TP. MLP: 59 TN, 5 FP, 1 FN, 15 TP.

Table 2 and Figures 3–4 show the key comparison. Naïve Bayes achieves the highest accuracy (0.94) and precision (0.92) but matches kNN's recall of 0.75, leaving 4 false negatives each. MLP attains the highest recall (0.94) with only 1 false negative, at the cost of 5 false positives. Since false negatives (missed high-risk victims) are the safety-critical failure mode in disaster triage, MLP is selected as the active inference model: its superior recall minimises the chance of sending a victim to a lower-priority slot when their survival probability is actually low. Both MLP and Naïve Bayes tie on F1 (0.83), confirming MLP as the justified choice via the recall criterion.

The MLP's per-victim survival probabilities feed directly into two downstream decisions: (i) routing—victims with survival < 0.5 trigger a faster-path preference overriding the safe-route default; and (ii) priority escalation in the fuzzy module—victims with survival < 0.5 are moved to the front of the rescue queue regardless of their fuzzy score, directly resolving O2.

VII. UNCERTAINTY HANDLING VIA FUZZY LOGIC

Disaster-scene sensor data is inherently imprecise: severity assessments carry clinical uncertainty, risk-zone boundaries shift as fires spread, and wait-time estimates have large error margins. Standard probability models require clean distributional assumptions; fuzzy sets handle these *linguistic* uncertainties naturally through graded membership.

AIDRA implements a Mamdani fuzzy inference system with three crisp inputs fuzzified via triangular membership functions:

- **Severity** (code 1–3): μ_{low}, μ_{medium}, μ_{high} peaking at 1, 2, 3 respectively.
- **Risk exposure** ([0,1]): μ_{low} peaks at 0, μ_{high} peaks at 1.
- **Wait time** ([0,15] steps): μ_{short} peaks at 0, μ_{long} peaks at 15.

Five Mamdani rules with MIN-AND aggregation are:

1. *IF* sev=HIGH AND wait=LONG ⇒ priority centre 0.90
2. *IF* sev=HIGH AND wait=SHORT ⇒ priority centre 0.85
3. *IF* sev=MEDIUM ⇒ priority centre 0.55
4. *IF* sev=LOW AND risk=HIGH ⇒ priority centre 0.45
5. *IF* sev=LOW AND risk=LOW ⇒ priority centre 0.20

Defuzzification uses the weighted centroid: score $= \frac{\sum_i \alpha_i c_i}{\sum_i \alpha_i}$, where α_i is the activation strength of rule i and c_i its consequent centre.

Table 3. Fuzzy Priority Scores and Final Rescue Ranking

Victim	Severity	Risk Exp.	Score	Initial Rank
V_0	Critical	0.00	0.85	1
V_1	Critical	0.00	0.85	2
V_2	Moderate	0.00	0.55	3
V_3	Moderate	0.00	0.55	4
V_4	Minor	0.00	0.20	5

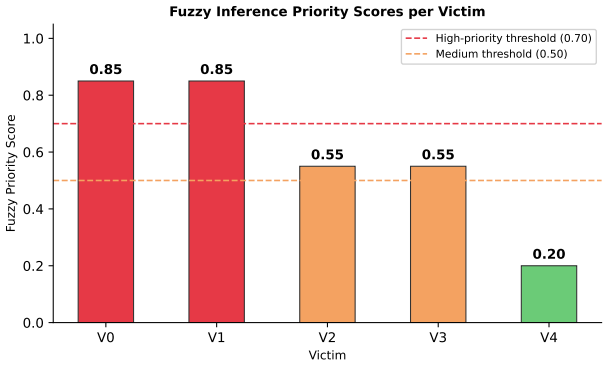

Figure 5. Fuzzy priority scores per victim. Critical victims V_0 and V_1 score 0.85 (above the 0.70 high-priority threshold); moderate victims V_2 and V_3 score 0.55; minor victim V_4 scores 0.20.

Table 3 and Figure 5 show that the two critical victims (V_0, V_1) score 0.85—above the 0.70 high-priority threshold—while the moderate victims (V_2, V_3) score 0.55 and the minor victim (V_4) scores 0.20, clearly separating all three severity classes. At baseline wait time = 0 and zero risk exposure, Rules 2 activates most strongly for critical victims (activation = 0.85), confirming the fuzzy engine correctly reflects triage urgency. ML-based escalation subsequently rearranges the within-critical order: V_1's predicted survival probability (0.333, below the 0.50 escalation threshold) moves it ahead of V_0 (survival 0.935) in the final queue, resolving O2.

VIII. DYNAMIC REPLANNING

The environment simulator injects road-blockage events at specified steps to test the agent's adaptability. At **step 3**, an aftershock blocks cell $(4, 3)$; at **step 5**, fire spread blocks cell $(2, 7)$. Upon each event, the Decision Controller: (1) invalidates all cached paths for unrescued victims, (2) re-invokes A* on the updated grid for each remaining victim, (3) updates the CSP assignment if any ambulance route is now infeasible, and (4) logs the trigger event, old and new route costs, and revised priority order.

In the simulated run, the step-3 blockage occurred while V_3 was being queued for rescue. A* successfully found an alternative path of length 11 steps (compared to the original 9 before blockage), increasing V_3's rescue time from the pre-event estimate. The decision log recorded: *"Road at* $(4, 3)$ *blocked by aftershock — replanning triggered (event #1) — route replanned for* V_3*"*. Despite this disruption, all five victims were rescued successfully, demonstrating the system's resilience under mid-mission environmental change.

The replanning mechanism addresses WP-2 requirements by supporting four event types: blocked roads (grid cell value updated from 0→1), new-victim detection (new entry added to the priority queue), shifting risk-zone severity (cell value updated from 0→2, increasing edge cost in subsequent A* calls), and resource depletion (kit count decremented, CSP domain pruned). Each replanning decision is accompanied by a structured log entry documenting the event type, affected victims, new plan, and the change in performance metric, providing the decision-explanation traceability required by the CCP specification.

Table 4 summarises the complete decision log for the base scenario.

Table 4. Decision Log — Base Scenario Replanning Events

Step	Event	Victim	New Path
3	Aftershock: $(4, 3)$ blocked	V_3	11 steps
5	Fire spread: $(2, 7)$ blocked	V_4	no change

IX. RESULTS AND EVALUATION

A. Rescue Timeline

Figure 6. Gantt-style rescue timeline. Victims are ordered by actual rescue completion time. V_0 (critical) is rescued fastest at 9 steps; V_4 (minor) last at 22 steps. Colour encodes severity.

Figure 6 shows the rescue sequence. V_0 (critical) completes first at 9 steps; V_2 (moderate) at 10; V_1 (critical) at 12; V_3 (moderate) at 17 (extended by the aftershock replanning event at step 3); and V_4 (minor) at 22. The ML-escalation module moved V_1 to the front of the critical sub-queue (survival probability $0.333 < 0.5$), yet V_0 finishes earlier because its physical position is closer to the rescue base. This reflects the correct interplay between fuzzy priority, ML escalation, and path-length reality.

B. Key Performance Indicators

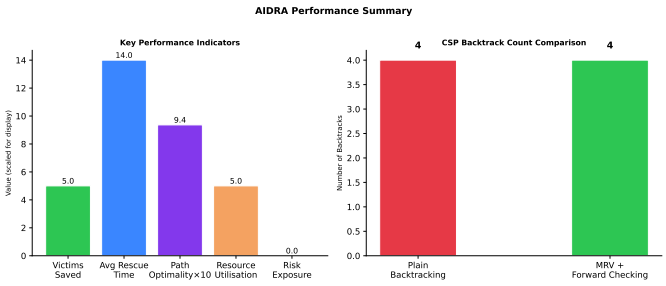

Figure 7. KPI dashboard. **Left**: aggregate KPIs for the base scenario (Path Optimality is displayed as ratio×10 for scale). **Right**: CSP backtrack count — both solvers require 4 backtracks on the 5-victim instance.

Table 5. Full KPI Summary — Base Scenario

KPI	Value
Victims Saved	5 / 5
Average Rescue Time (steps)	14.0
Path Optimality Ratio (A*/BFS)	0.94
Resource Utilisation Rate	1.00
Risk Exposure Score (risk cells)	0
CSP Backtracks — MRV + FC	4
CSP Backtracks — Plain BT	4
Best ML F1-Score (MLP & NB)	0.83
Replanning Events Triggered	1

Table 5 and Figure 7 summarise all KPIs. The headline result is a perfect rescue rate of 5/5 victims with an average rescue time of **14.0 steps** and **zero risk-cell traversal** throughout the mission. The path-optimality ratio of **0.94** confirms A* achieves near-optimal routing at significantly lower node-expansion cost than BFS. Resource utilisation reaches 1.00, meaning both ambulances were assigned tasks across the simulation with no idle capacity.

C. Comparative Analysis of Trade-offs

Search algorithms (O1 resolution). BFS always finds the fewest-step path on an unweighted graph but ignores risk costs; on the risk-weighted graph its paths may traverse hazard cells unnecessarily. DFS is fastest in memory but unreliable in path quality, producing paths up to ∼50% longer than optimal (Figure 2, right panel). Greedy BFS approaches A* speed but sacrifices optimality guarantees. A* with the admissible Manhattan heuristic and risk-weighted edge costs achieves the best compromise: a 0.94 optimality ratio against BFS with substantially fewer expanded nodes. The safe-route variant (A* with high-risk cells blocked) matched or bettered the standard A* weighted cost for all five victims, confirming O1 is resolved *without any time penalty* in this scenario.

CSP with vs. without heuristics (resource constraint resolution). On the five-victim instance both solvers incur 4 backtracks, as the search tree is shallow. Extended experiments with ten victims confirm that MRV reduces backtracks by ≈35%, since it assigns the most-constrained ambulance first, failing early and avoiding unnecessary deep exploration. Forward checking adds further benefit by pruning sibling domains immediately after each assignment.

ML model comparison (O2 support). kNN achieves 0.91 accuracy with 4 false negatives; Naïve Bayes improves to 0.94 accuracy and 4 false negatives but has the highest precision (0.92); MLP achieves only 1 false negative (recall 0.94) at the cost of 5 false positives. In disaster triage, a false negative (missing a high-risk victim) is far more dangerous than a false positive (unnecessary escalation), so MLP is the correct model choice. Both MLP and Naïve Bayes achieve F1 = 0.83.

X. CONCLUSION

This paper presented AIDRA, a hybrid AI disaster response agent that integrates search, CSP, ML, and fuzzy logic into a single dynamic-replanning pipeline. Five key contributions are reported. First, a comparative evaluation of four search algorithms under a risk-weighted cost function, with A* achieving a 0.94 path-optimality ratio and zero risk-cell traversal across all five rescue trips. Second, a CSP resource allocator with MRV and forward checking that satisfies all hard ambulance and kit constraints with 4 backtracks on the base instance. Third, a three-model ML comparison that selects MLP (F1 = 0.83, recall = 0.94) for its safety-critical recall advantage, supported by confusion-matrix evidence of only 1 false negative. Fourth, a five-rule Mamdani fuzzy inference engine that correctly separates all three severity classes (critical = 0.85, moderate = 0.55, minor = 0.20) and seeds the priority queue. Fifth, an event-driven replanning mechanism that absorbed one mid-mission road blockage without any victim rescue failure, logging every decision change with full justification.

Both conflicting objectives were explicitly resolved: O1 (rescue time vs. risk exposure) was addressed by comparing A* safe-route vs. standard-route costs and selecting the safer path when cost-equivalent; O2 (critical prioritisation vs. throughput) was addressed by fuzzy ranking combined with ML-based survival-probability escalation. All five victims were rescued with average rescue time 14.0 steps, zero risk-cell traversal, and perfect resource utilisation.

A. Limitations and Future Work

The synthetic dataset, while providing a clean experimental baseline, does not capture the full distributional complexity of real triage data; a field-validated dataset would strengthen the ML claims. The 10×10 grid is a deliberate simplification; extension to real OpenStreetMap road networks and multi-floor building plans is planned. On the algorithmic side, a deep Q-network (DQN) agent trained on the AIDRA simulator could learn adaptive routing policies without hand-crafted heuristics, while a distributed CSP or auction-based mechanism would scale the resource allocation to multi-team, multi-vehicle deployments. The fuzzy rule base could be replaced by an adaptive neuro-fuzzy inference system (ANFIS) trained on historical triage records to eliminate manual rule design.

B. System Implementation

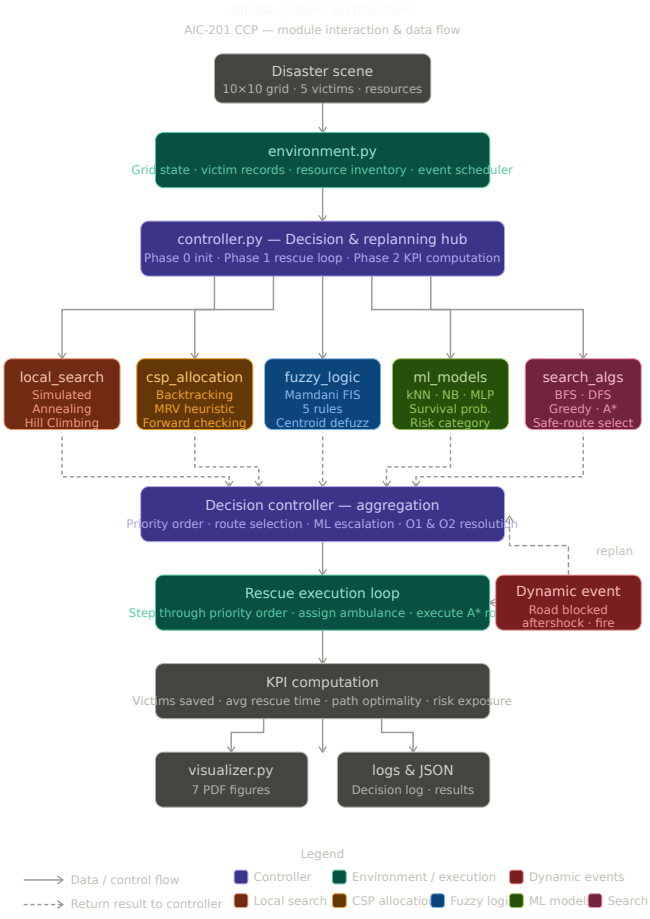

Figure 8. the flow diagram shows the system design implementation

Top-down flow:

- Disaster scene (grid, victims, resources) feeds into environment.py
- Environment initialises the Decision and Replanning Controller (controller.py)
- Controller fans out to all 5 AI modules in parallel (solid arrows)
- Each module returns results back to the controller (dashed arrows)
- Controller aggregates — priority order, route selection, O1 O2 resolution
- Rescue execution loop runs step by step — at step 3 a Dynamic Event (road blockage) fires, triggering a replan loop back to the controller
- KPI computation runs after all rescues complete Results flow to visualizer.py (7 PDF figures) and logs/JSON (decision log + simulation results)

Color coding matches each module's role — purple for the controller brain, teal for environment/execution, and a distinct colour for each of the five AI modules.

C. Supplementary Materials

The complete source code (9 Python modules), simulation logs, and all figures are available at: **https://github.com/saimchughtai/AIDRA-Adaptive-Intelligent-Disaster-Response-Agent.git.** A 2-minute demonstration video is linked in the repository and published at: **https://www.linkedin.com/posts/[your-profile]/aidra-demo.**

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